

ToricGT: Compact Graph-Token Reasoning with Tropical Ring Attention, Toric Diagnostics, and Random-Order Compression

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Abstract

We present ToricGT, a compact graph-to-graph Transformer family for algebraic, mathematical, morphological, and parameter-constrained language-model reasoning. The research model tokenizes nodes and edges as a shared sequence, uses exact blockwise ring evaluation for tropical max-plus attention, adds finite noncommutative-torus phase channels, and uses permutation-equivariant Soft-MoE routing in upper graph blocks. The paper contributes a finite theory and an implementation-facing training contract. We prove graph-token equivariance, exactness of tropical ring attention, stability of active faces under perturbation, and simulation of bounded semiring dynamic programs. Motivated by recent toric geometry of ReLU networks, we further interpret tropical head decisions as active faces of Newton polytopes and normal fans, giving compact diagnostics for margins, bends, and future toric auxiliary losses. A separate OpenAI Parameter-Golf track projects byte chunks to random-order graph positions and trains a dense recurrent micro language model with tied embeddings, toric position phases, tropical-ring attention, and byte-accounted int8 export. The result is a small, auditable model design whose reasoning behavior is inspected through equivariance errors, active faces, toric moments, artifact bytes, BPB, and GFlowNet diversity rather than scalar accuracy alone.

1 Introduction

Many reasoning tasks are naturally graph-to-graph transformations: proof steps depend on earlier equations, graph algorithms update node and edge states, Hebrew morphology links surface forms to roots and templates, and algebraic words reduce through local rewrite relations. ToricGT targets this regime with three design constraints. First, relabeling a graph should relabel node and edge outputs, not change the computation. Second, dynamic-programming-like reasoning should be visible through max-plus active faces and margins. Third, compactness matters: the same ideas should support a small research model and a Parameter-Golf-style byte model under a strict artifact budget.

The main novelty is the integration of algebraic structure with trainable graph-token Transformers. Tropical attention implements max-plus gates and exposes provenance. Ring evaluation changes the schedule but not the function. Finite Weyl pairs encode projective toric phase memory. Soft-MoE gives typed differentiable routing in the research graph model. For Parameter Golf, the same graph-token idea is compressed into random-order autoregressive decoding: each byte position becomes a graph node scored in a content-independent random order. Finally, toric geometry supplies a new diagnostic layer: active candidates in tropical heads form exposed faces of Newton polytopes, and changes across cell walls behave like bend or Cartier-divisor intersection measurements.

2 Model

A ToricGT instance is

$$F(G) = D \circ \Phi_L \circ \dots \circ \Phi_1 \circ T(G), \quad (1)$$

where T maps a typed attributed graph to node and edge tokens, Φ_ℓ is a shared token-equivariant block, and D is a shared node/edge/graph decoder. A graph is

$$G = (V, E, s, t, \tau_V, \tau_E, x, e, m_V, m_E), \quad (2)$$

with source and target maps for edge tokens, finite node and edge types, bounded attributes, and padding masks.

Tropical ring attention. For score matrix S and values V , a tropical head computes

$$Y_{ic} = \max_j (S_{ij} + V_{jc}), \quad A_{ic} = \arg \max_j (S_{ij} + V_{jc}). \quad (3)$$

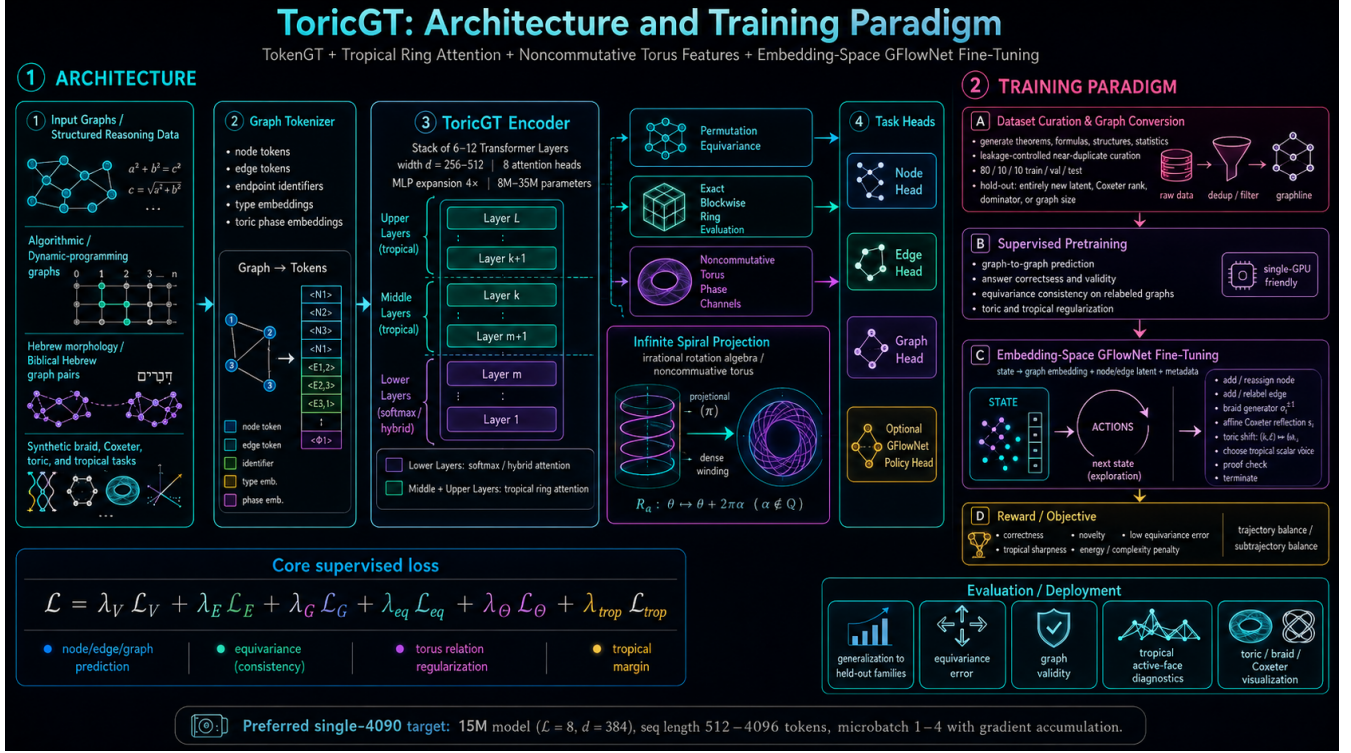


Figure 1: ToricGT combines graph tokenization, tropical ring attention, noncommutative-torus phase channels, Soft-MoE routing, and embedding-space GFlowNet fine-tuning.

Partitioning keys into blocks $B_1, \dots, B_R$ gives

$$\max_j (S_{ij} + V_{jc}) = \max_r \max_{j \in B_r} (S_{ij} + V_{jc}), \quad (4)$$

so ring evaluation is exactly equal to full tropical attention in exact arithmetic.

Soft-MoE routing. For hidden states $X \in \mathbb{R}^{B \times L \times d}$, learned slot vectors $\Phi \in \mathbb{R}^{d \times E \times S}$, and validity mask M , define

$$A_{bies} = g \langle x_{bi} / \|x_{bi}\|, \phi_{es} / \|\phi_{es}\| \rangle, \quad (5)$$

$$D_{bies} = \frac{\exp A_{bies} M_{bi}}{\sum_j \exp A_{bjes} M_{bj} + \epsilon}, \quad (6)$$

$$C_{bies} = \frac{\exp A_{bies}}{\sum_{e', s'} \exp A_{bie' s'} + \epsilon}, \quad (7)$$

$$Z_{bes} = \sum_i D_{bies} X_{bi}, \quad Y_{bi} = \sum_{e, s} C_{bies} f_e(Z_{bes}). \quad (8)$$

We use $E \geq 4$ experts by default in the graph model. The Parameter-Golf micro-LM is dense by default because a full expert bank is usually a poor bytes-per-quality tradeoff under a 16 MB artifact cap; prefix-causal or low-rank Soft-MoE remains an ablation after the dense baseline is stable.

3 Theory

Theorem 1 (Graph equivariance). *If tokenization, masks, attention projections, tropical heads, Soft-MoE routing, residuals, normalization, and decoders are shared over token rows and use only equivariant incidence/type information, then ToricGT is graph-to-graph equivariant.*

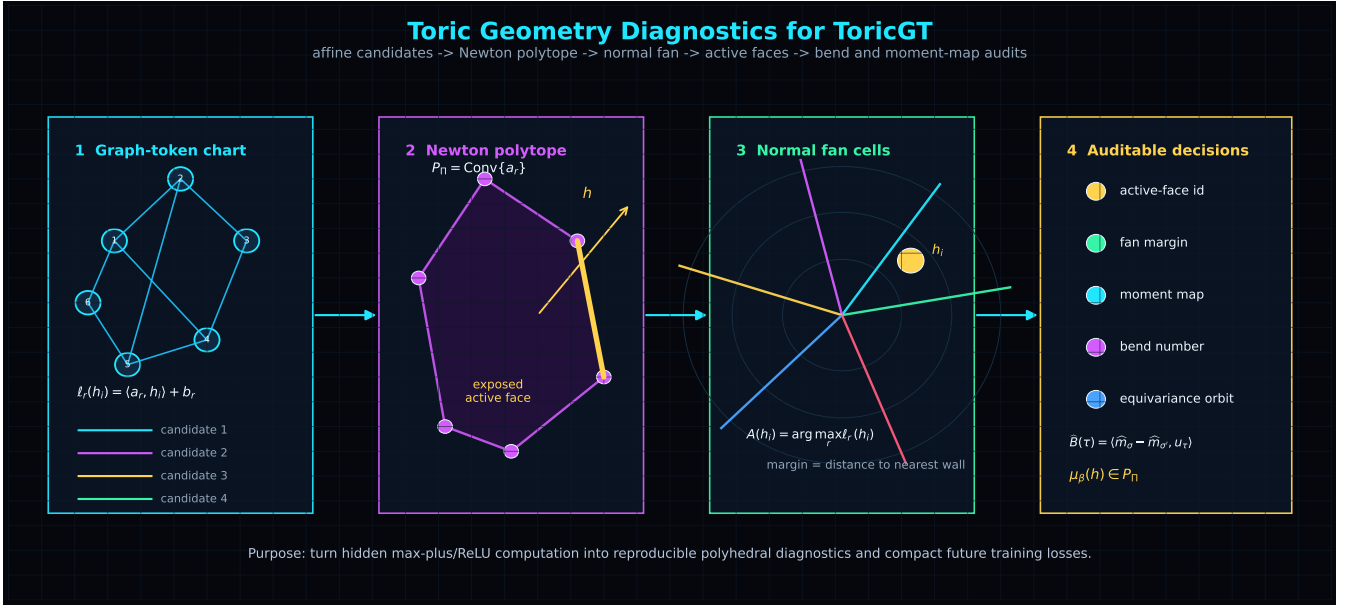


Figure 2: Toric diagnostics turn tropical candidate scores into Newton polytopes, normal-fan cells, active-face margins, and bend measurements.

Proof. Tokenization maps relabeled graphs to row-permuted token tables. Linear projections commute with row permutations. Softmax attention, tropical attention, and Soft-MoE dispatch/combination commute with the same row permutation because their normalizers sum over either all valid tokens or slots for a fixed token. Row-wise residual, normalization, and decoder maps commute with permutations. Composition preserves equivariance. $\square$

Lemma 2 (Active-face stability). *If $\|S - \hat{S}\|_\infty \leq \epsilon_S$ and $\|V - \hat{V}\|_\infty \leq \epsilon_V$, then tropical outputs obey*

$$\|Y - \hat{Y}\|_\infty \leq \epsilon_S + \epsilon_V. \quad (9)$$

If the top-two margin is larger than $2(\epsilon_S + \epsilon_V)$, the active key is unchanged.

Proof. Each candidate score changes by at most $\epsilon_S + \epsilon_V$, and the maximum is 1-Lipschitz in ℓ_∞ . The margin condition keeps the original maximizer strictly above all competitors after perturbation. $\square$

Toric diagnostic theorem. Let a tropical probe be $\psi(h) = \max_r (\langle a_r, h \rangle + b_r)$ with rational slopes a_r . Its Newton polytope is $P = \text{Conv}(a_1, \dots, a_R)$. The active set

$$A(h) = \arg \max_r (\langle a_r, h \rangle + b_r) \quad (10)$$

is the exposed face of the lifted polytope $\text{Conv}\{(a_r, b_r)\}_r$ under the linear functional $(a, b) \mapsto \langle a, h \rangle + b$. Hence tropical active-face diagnostics are cells of a normal fan. This imports the ReLU fan / Cartier-divisor viewpoint of Fu [1] into ToricGT tropical subcomputations, complementing earlier tropical rational interpretations of ReLU networks [2].

4 Training And Evaluation

The supervised objective combines node, edge, and graph losses with equivariance, torus, tropical-margin, Hebrew morphology, Soft-MoE, cache, and distillation terms:

$$\begin{aligned} \mathcal{L} = & \lambda_V \mathcal{L}_V + \lambda_E \mathcal{L}_E + \lambda_G \mathcal{L}_G + \lambda_{\text{eq}} \mathcal{L}_{\text{eq}} + \lambda_\Theta \mathcal{L}_\Theta + \lambda_{\text{trop}} \mathcal{L}_{\text{margin}} \\ & + \lambda_H \mathcal{L}_H + \lambda_{\text{moe}} \mathcal{L}_{\text{moe}} + \lambda_{\text{cache}} \mathcal{L}_{\text{PQ}} + \lambda_{\text{KD}} \mathcal{L}_{\text{distill}}. \end{aligned} \quad (11)$$

Embedding-space GFlowNet fine-tuning samples reasoning graphs with probability proportional to reward using trajectory balance. The reward combines correctness, novelty, low equivariance error, tropical margin, low toric commutator error, and complexity penalties.

The data mixture is graph-first: math reasoning, graph-of-thought traces, algorithmic tasks, synthetic rotation/tropical tasks, Hebrew morphology graphs, and Jewish Hebrew text sources are normalized into Parquet records with stable content hashes and graph JSON. Splits are clustered by near-duplicate text, graph sketches, algebraic normal forms, root families, and source references.

Metric	Failure mode exposed
Equivariance error	absolute-position leakage and edge-order bugs
Active-face margin	unstable dynamic-programming decisions
Toric bend consistency	incoherent polyhedral reasoning changes
Soft-MoE expert mass	expert collapse or over-routing
GFlowNet diversity	mode collapse and reward hacking
Parameter-Golf BPB/bytes	compactness and scoring compliance

5 Parameter-Golf Track

The contest-facing model is not the full graph encoder. It is a dense random-order byte model with tied embeddings, recurrent block reuse, toric target-position phases, lower softmax attention, upper tropical-ring attention, and byte-accounted int8 export. The artifact budget is

$$B_{\text{art}} = B_{\text{code}} + B_{\text{weights}} + B_{\text{tokenizer}} + B_{\text{metadata}} \leq 16,000,000. \quad (12)$$

For a byte chunk $b = (b_1, \dots, b_T)$, sample a content-independent order $\pi \in S_T$ from public metadata and factor

$$q_{\theta}(b \mid \pi) = \prod_{k=1}^T q_{\theta}(b_{\pi_k} \mid b_{\pi_{<k}}, \pi_{\leq k}). \quad (13)$$

BPB is computed prequentially: predict a normalized next-byte distribution, score the observed byte, then update the state. Because π is independent of unscored byte values and row k never contains b_{π_k} , random-order decoding is score-before-update valid while exposing more graph-like order diversity than ordinary left-to-right decoding. Dense recurrence is the default because it adds computation without adding stored expert weights; Soft-MoE adapters are retained only as byte-accounted ablations.

6 Next Iteration

The next ToricGT iteration should use toric geometry as a training signal. First, freeze the trained checkpoint and build empirical toric shadows: occupied fan cells, fitted slopes, active-face margins, and bend magnitudes. Second, add synthetic toric tasks: Newton-polytope active-face prediction, Cartier-divisor bend prediction, toric ideal relation checks, and ReLU fan realization tests. Third, attach low-rank quantized toric probes and optimize

$$\mathcal{L}_{\text{toric}} = \mathcal{L}_{\text{base}} + \lambda_{\text{fan}} \mathcal{L}_{\text{fan}} + \lambda_{\text{bend}} \mathcal{L}_{\text{bend}} + \lambda_{\text{binom}} \mathcal{L}_{\text{binom}} + \lambda_{\text{moment}} \|\hat{\mu} - \mu^*\|_2^2. \quad (14)$$

Finally, make inference-time scaling geometric: continuous GFlowNets can move inside normal cones, along walls, and across moment polytopes while discrete actions edit graph-of-thought states. Continuous GFlowNet theory supplies the density-correct trajectory-balance objective for these hybrid paths [7]. This keeps the architecture small while giving search a structured latent space.

7 Conclusion

ToricGT is a finite, auditable model family. Its exact claims are graph-token equivariance, ring-schedule exactness, max-plus semiring simulation on bounded domains, projective toric phase consistency, random-order score validity for byte modeling, and perturbation bounds for compressed caches. Its practical claim is that reasoning models should expose the geometry of their decisions. Tropical active faces, Newton polytopes, toric bends, artifact bytes, BPB, expert utilization, and GFlowNet reward landscapes make small-model reasoning easier to test, compress, and improve.

References

- [1] Yaoying Fu. Toric geometry of ReLU neural networks. *arXiv:2509.05894*, 2025.
- [2] Liwen Zhang, Gregory Naitzat, and Lek-Heng Lim. Tropical geometry of deep neural networks. In *ICML*, PMLR 80:5824–5832, 2018.
- [3] Jinwoo Kim, Tien Dat Nguyen, Seonwoo Min, Sungjun Cho, Moontae Lee, Honglak Lee, and Seunghoon Hong. Pure Transformers are powerful graph learners. *NeurIPS*, 2022.
- [4] Hao Liu, Matei Zaharia, and Pieter Abbeel. Ring Attention with blockwise Transformers for near-infinite context. *arXiv:2310.01889*, 2023.
- [5] Joan Puigcerver, Carlos Riquelme, Basil Mustafa, and Neil Houlsby. From Sparse to Soft Mixtures of Experts. *ICLR*, 2024.
- [6] Yoshua Bengio, Salem Lahlou, Tristan Deleu, Edward J. Hu, Mo Tiwari, and Emmanuel Bengio. GFlowNet foundations. *JMLR*, 24(210):1–55, 2023.
- [7] Salem Lahlou, Tristan Deleu, Pablo Lemos, Dinghuai Zhang, Alexandra Volokhova, Alex Hernández-García, Lena Nehale Ezzine, Yoshua Bengio, and Nikolay Malkin. A theory of continuous generative flow networks. In *ICML*, PMLR 202:18269–18300, 2023.
- [8] Insu Han, Praneeth Kacham, Amin Karbasi, Vahab Mirrokni, and Amir Zandieh. PolarQuant: Quantizing KV Caches with Polar Transformation. *arXiv:2502.02617*, 2025.
- [9] Marie-Charlotte Brandenburg, Georg Loho, and Guido Montufar. The real tropical geometry of neural networks. *arXiv:2403.11871*, 2024.